



MechaMix: The Automated Drink-Mixing Machine

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What is the Problem?

01

Human Inconsistency

Manual bartending results in volumetric variance, leading to inconsistent taste and inventory waste.

02

Machines Lack Precision

Current machines lack dispensing capabilities for high-viscosity liquids due to reliance on time-based pouring.

03

Hygiene & Labor Cost

High-traffic venues face high labor costs, slow service, and significant sanitation challenges.

Key Objectives & Specifications

± 0.1 mL

Volumetric Accuracy

$5 \pm 2^\circ\text{C}$

Serving Temperature

< 60 s

Prep-to-Serve Time

< \$800

Total Build Budget

5 Channels

3 Adaptive + 2 Fixed Ingredient Reservoirs

45–250 mL

Drink Volume Range

Controls & Firmware

Non-Blocking State Machine

IDLE → SELECT → DISPENSE → MIX → SERVE → CLEAN via step-index + timestamp tracking. Zero delay() calls throughout.

Flash Persistence

Pour volumes, ratios, and clean durations persist across power cycles. Safety limits are hardcoded and not user-accessible.

Emergency Stop

Cuts power to all pump motors and seals the dispense valve. Triggered from the touchscreen.

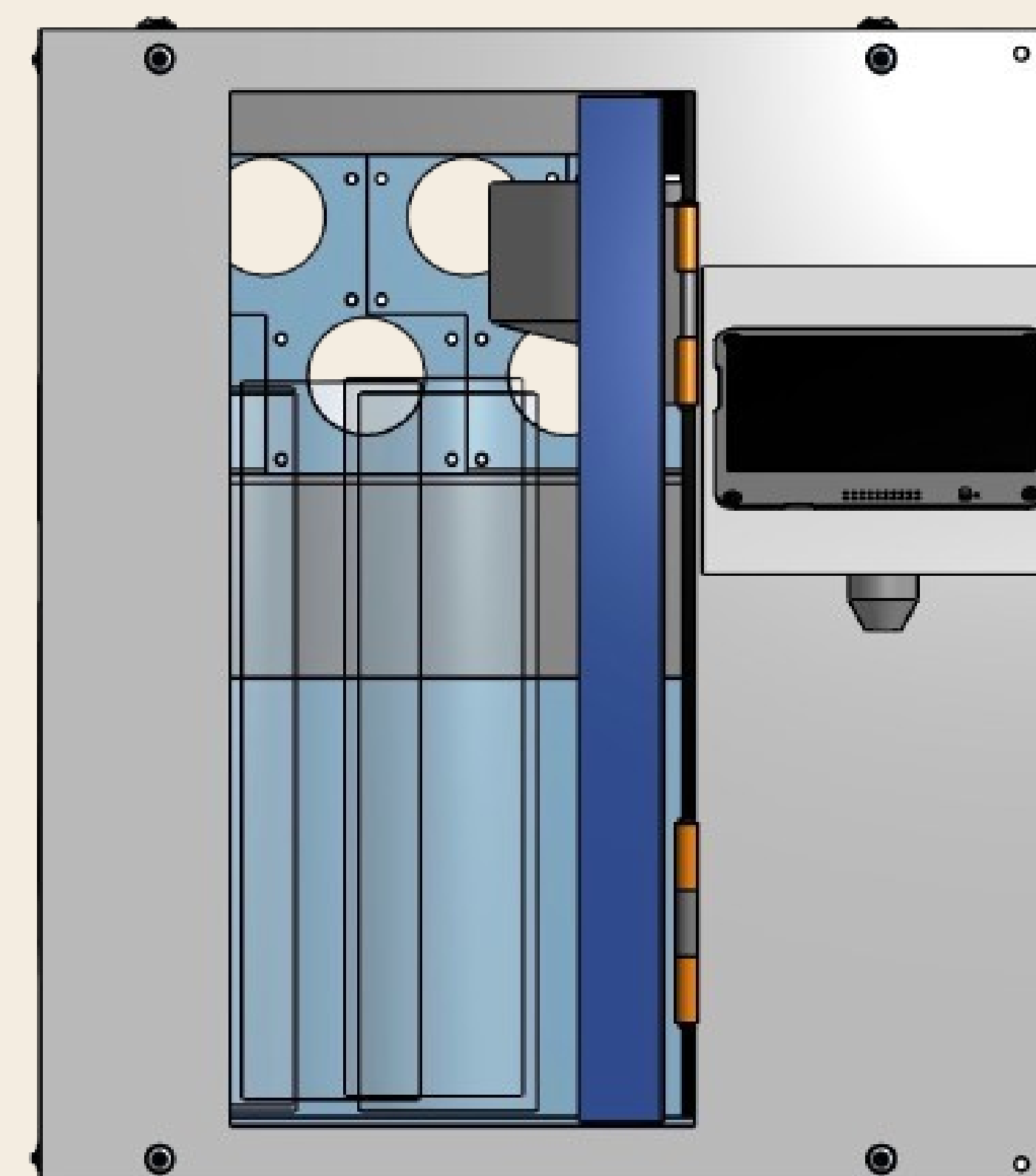
WiFi REST API

Edit recipes from browser on the local network. No physical access required.

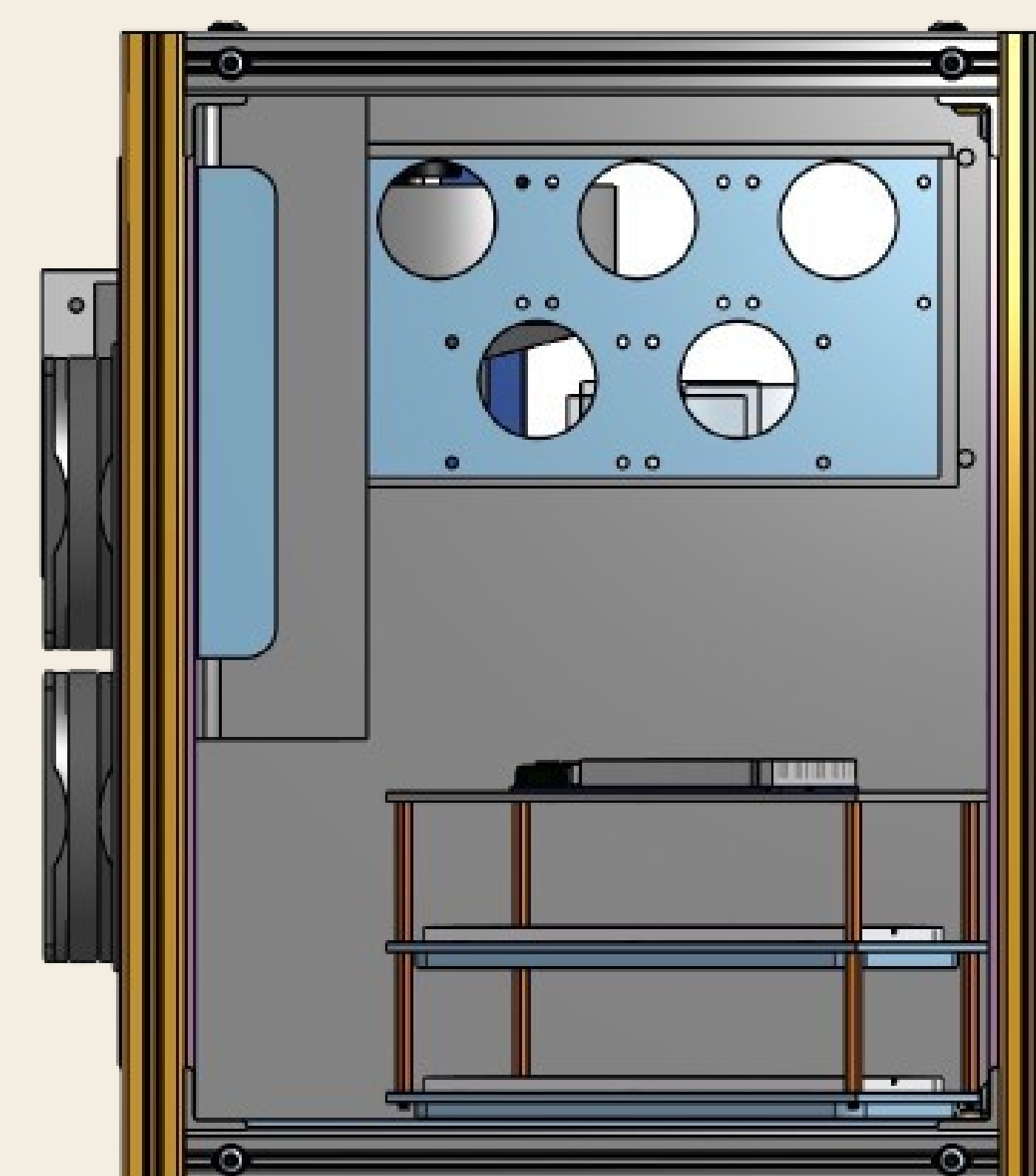


MechaMix Full Assembly | $0.5 \times 0.35 \times 0.5 \text{ m}^3$ • ~12 kg • Aluminum T-Slot Frame

Cooling Cavity



Electronic Cavity



Design & Engineering

Volume-Based Pump Control

Each pump is calibrated in mL/revolution, thus eliminating viscosity error entirely across all ingredients.

V-Bottom Cylindrical Chamber

Cylindrical V-bottom, yielding homogeneous blends with less rinse water.

NC Solenoid Fail-Safe

Nozzle valve is sealed by default. Power loss or emergency stop triggers immediate closure reducing spill risk.

FDA-Compliant Isolation

All electronics sealed in its own chamber, separated from food-contact surfaces. No cross-contamination pathway exists.

Insulated Reservoir Chamber

Reservoirs sit in an acrylic interior chamber sealed with insulation. 20mm of insulation minimizes thermal load on the Peltier.

Thermal Management

144W Peltier Cooler maintains temperature at $5 \pm 2^\circ\text{C}$ during operation. Pre-chilled ingredients are recommended to reduce the cooling burden.

Modular Reservoir Design

Three adaptive slots and two fixed slots handle all five ingredients, preventing cross-contamination between channels.

Structural Frame

Aluminum T-slot extrusions provide rigidity. Panels and brackets are 3D-printed PLA with acrylic walls.

Testing & Validation

Round 1

Steady-State Cooling

Threshold: $5 \pm 2^\circ\text{C}$

Volumetric Accuracy

Threshold: $\pm 0.1 \text{ mL}$

Pinch Valve Seal

Threshold: 0 drops/min

Round 2

Mixing Homogeneity

Threshold: Visually Uniform

Dynamic Seal Integrity

Threshold: No Leaking

Round 3

Cleaning Efficiency

Threshold: Visually Clear

Dispense Cycle Time

Threshold: < 60 seconds

Structural Rigidity

Threshold: Motionless

The Drink Journey: How MechaMix Works

Power On & Pre-Chill

Peltier Cooling

Cooler activates. Temperature Sensor holds chamber at $5 \pm 2^\circ\text{C}$.

1

2

Select Recipe

Touchscreen UI

Browse preset & custom recipes on the display.

Dispense Ingredients

Peristaltic Pumps

Pumps sequence in order, delivering at $\pm 0.1 \text{ mL}$ accuracy.

3

4

Blend & Mix

V-Bottom Chamber

Metal mixing rod runs for 10s of blending.

Serve Drink

NC Solenoid Valve

Normally-closed valve opens to dispense final beverage.

5

6

Auto Clean

Quick + Deep Modes

Quick Clean: flush + high-RPM spin, <50 mL, <2 min..